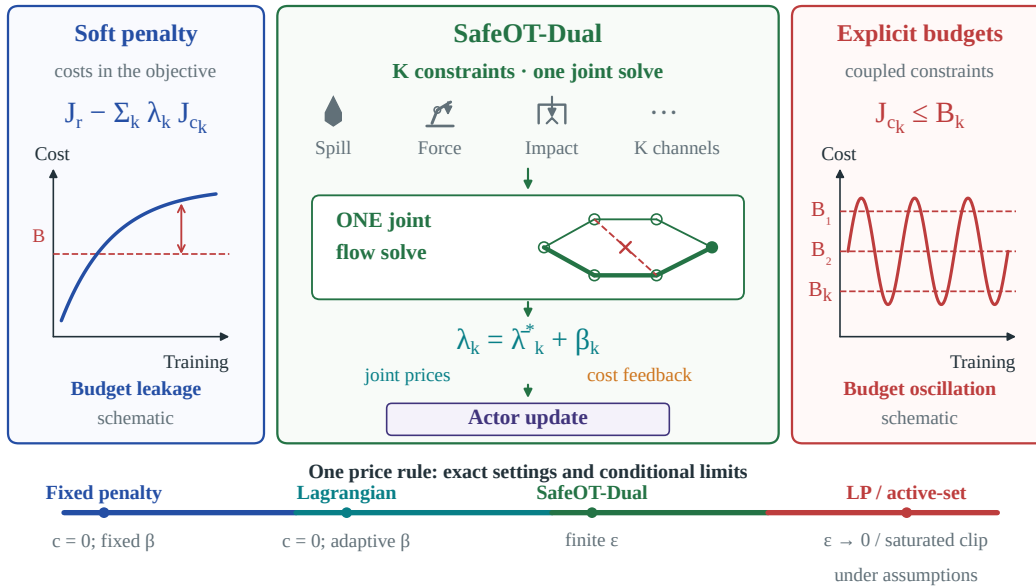




SafeOT-Dual unifies penalty and budget views through one joint flow solve. Soft budgets enter jointly; hard events remove graph edges. Clipped, filtered dual prices combine with realized-cost feedback before the actor update. Mini-plots illustrate possible leakage and oscillation, not experimental traces. Fixed penalties and Lagrangians are exact settings at $c = 0$; LP and saturated-clip endpoints are conditional limits.